

REAL-TIME AI -AWS CLOUD SECURITY MONITORING

AWS DevOps Internship Project — AP ANNAPUYAM · Batch 140



AWS

Cloud-native
infrastructure and
managed services



AI

Automated
analysis &
recommendations



SOC

Real-time security
operations



Network

Campus visibility &
telemetry

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Project Overview

Problem

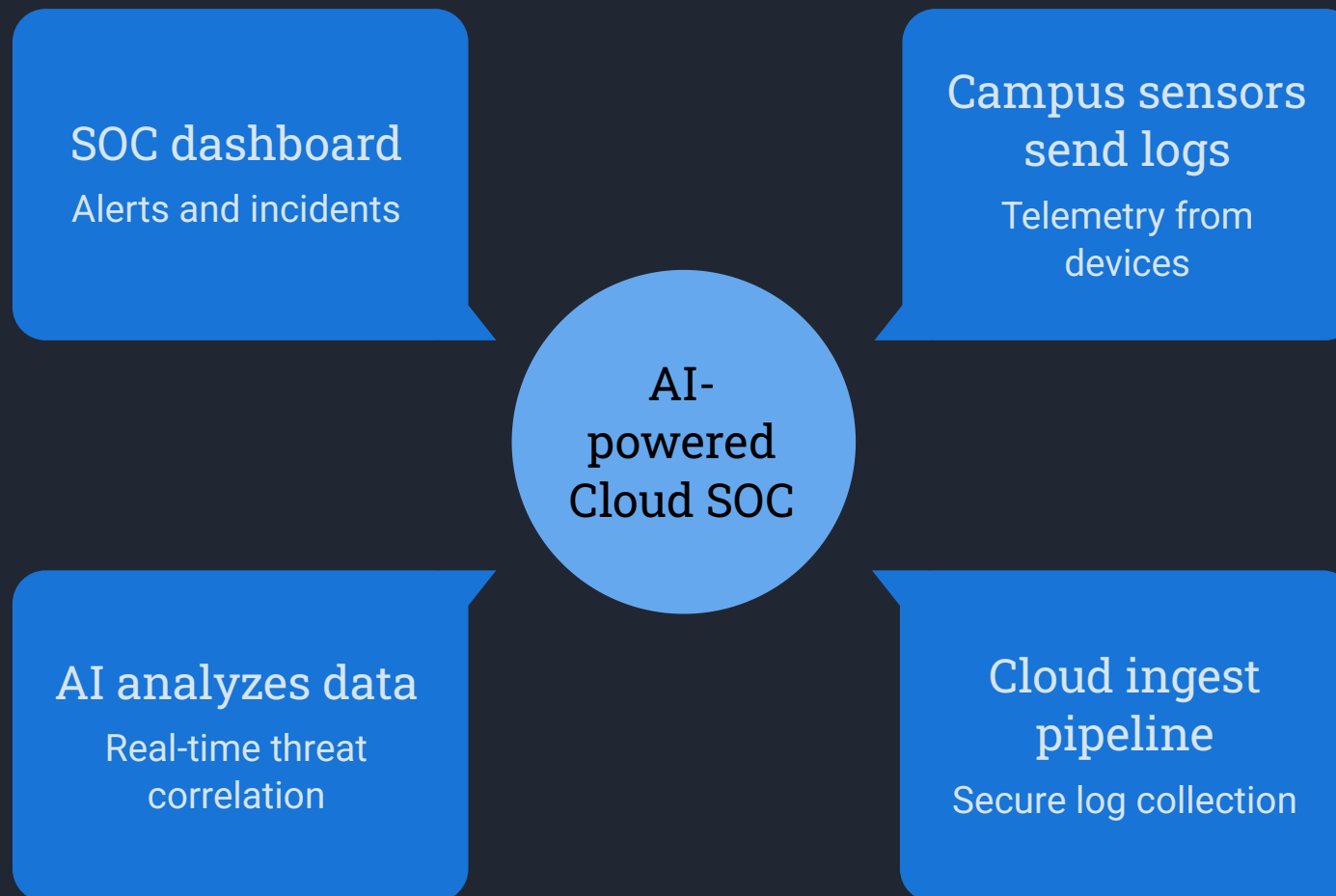
Traditional campus monitoring cannot detect, correlate, or respond to threats in real time — blind spots and delayed response increase risk.

Solution

An AI-powered cloud SOC that ingests AWS logs and campus telemetry to detect threats, generate risk scores, create incidents, and recommend actions automatically.

Objectives

- Real-time monitoring
- Automated threat detection
- AI analysis & risk scoring
- Incident management & reporting



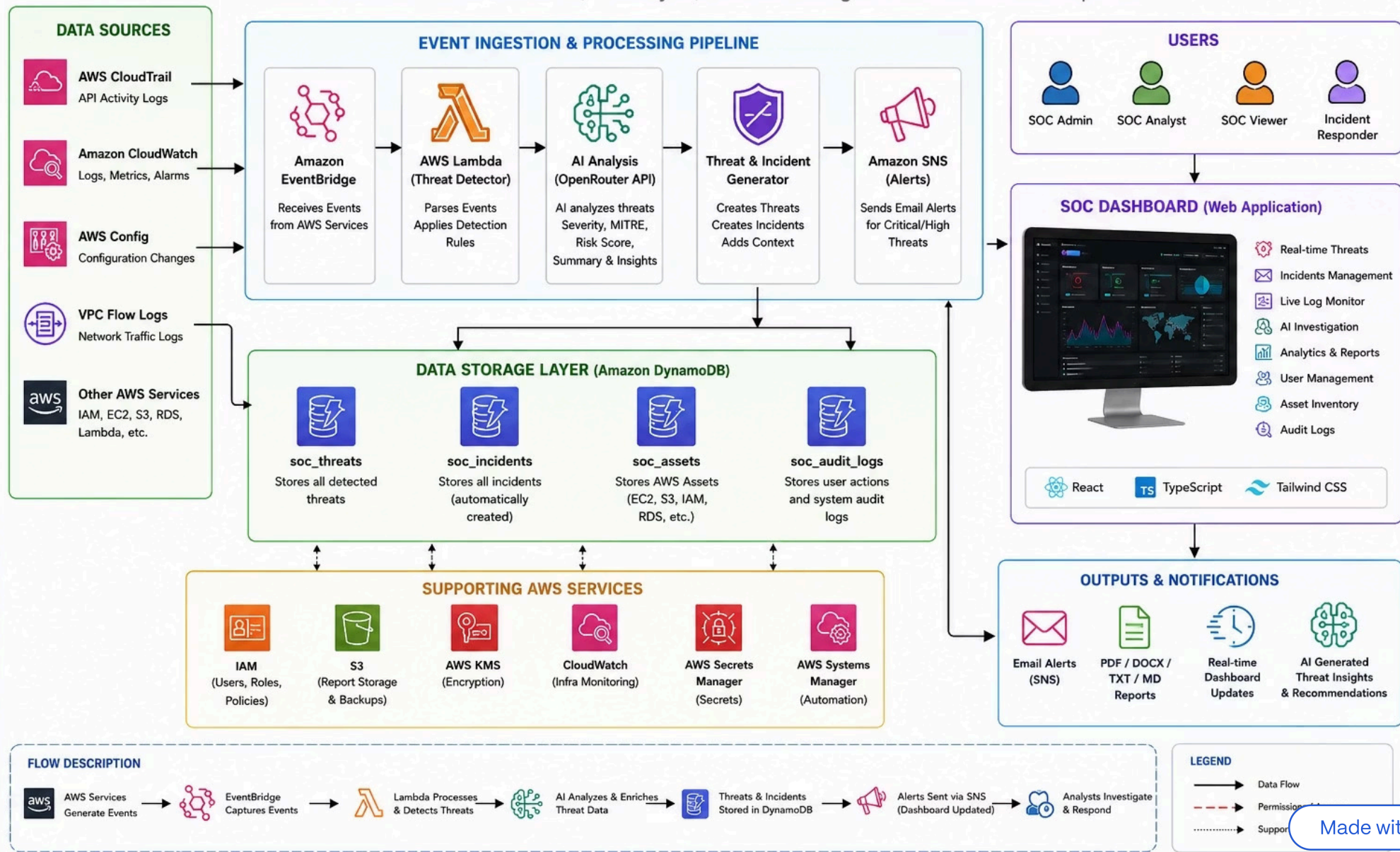
Large infographic shows end-to-end flow, icons for campus, cloud, AI, dashboard and reports.



AWS Cloud

AI-POWERED AWS CLOUD SECURITY OPERATIONS CENTER (SOC)

Real-time Threat Detection, AI Analysis, Incident Management & Automated Response



Technologies Used



Frontend

React · TypeScript ·
Tailwind CSS · Vite



Backend

Python · FastAPI ·
JWT Authentication



AWS Services

CloudTrail ·
EventBridge ·
Lambda · DynamoDB
· SNS · IAM · EC2



Deployment

Ubuntu · Nginx ·
GitHub Actions ·
Container-ready



AI

OpenRouter AI · LLM
analysis · Automated
recommendations



Storage & Infra

DynamoDB for
events, S3 (reports),
IAM controls

Icons: React, Python, FastAPI, DynamoDB, Lambda, CloudTrail, EC2, GitHub

Dashboard Features

Real-time Threat Dashboard

Live events, prioritized alerts, and contextual details

Threat Severity & Timeline

Risk scoring, time-series event view, escalation markers

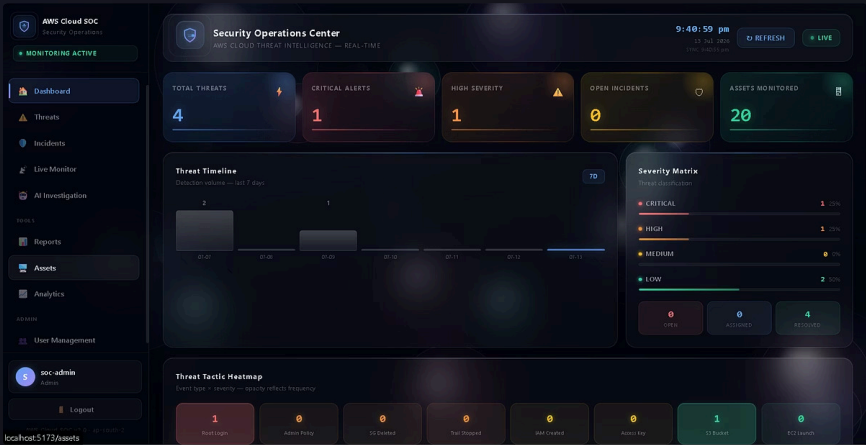
Asset Monitoring

Inventory, baselines, asset risk exposure

AI Recommendations

Actionable remediation steps and playbook links

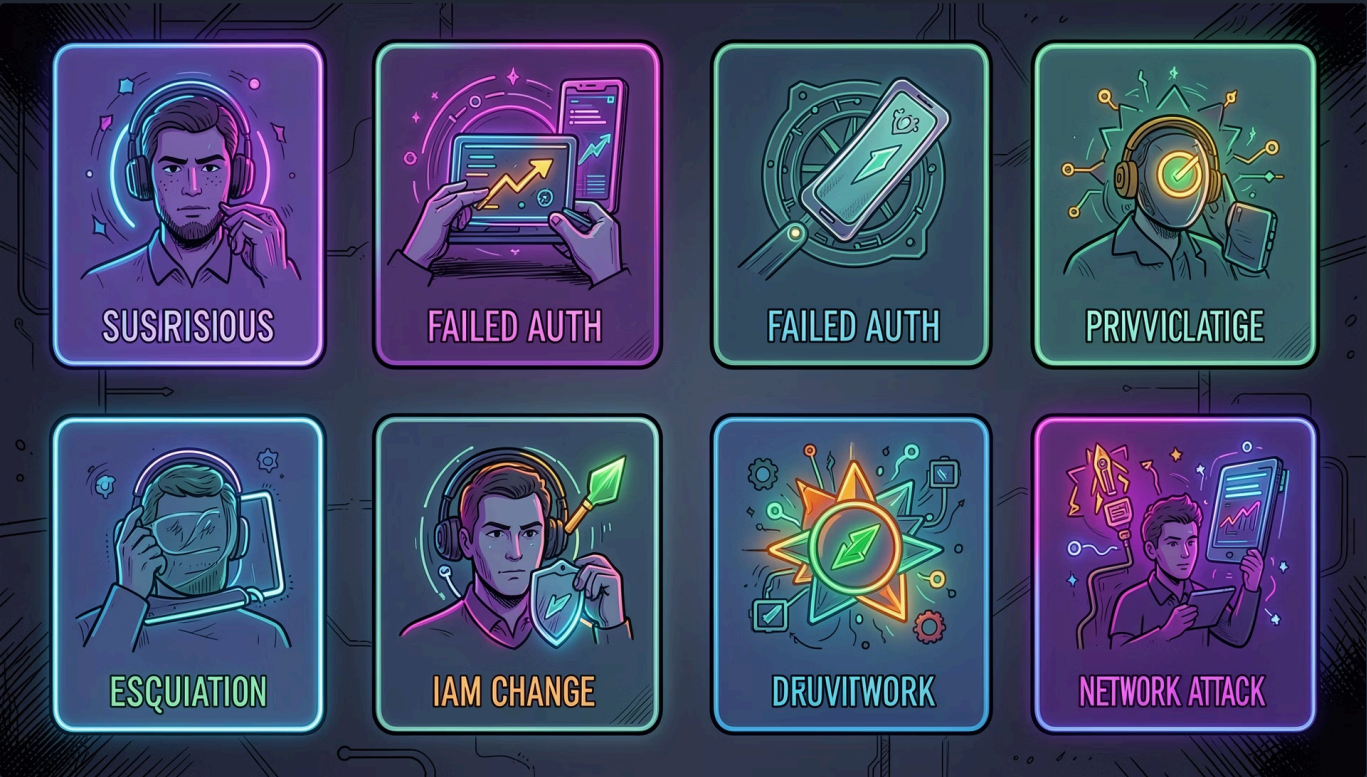
Role-based views, live filtering, and exportable incident summaries.



AI Threat Analysis



CloudTrail Threat Detection AI Analysis Risk Score Recommendations



1

Detected Patterns

Root login, suspicious login patterns, failed authentication bursts, privilege escalation attempts, IAM policy changes, network scanning indicators.

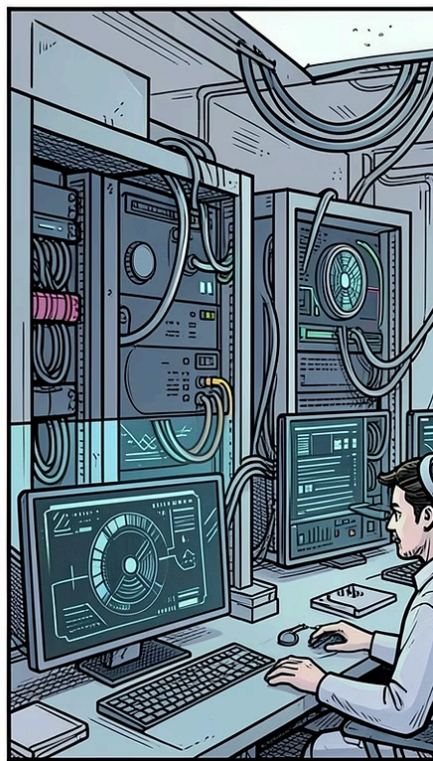
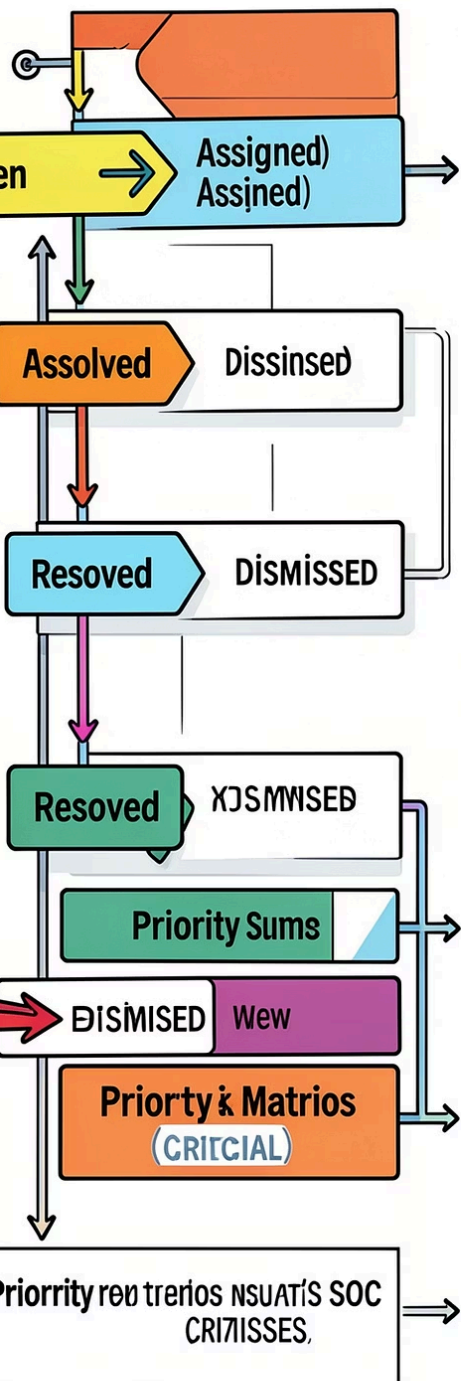
2

AI Outputs

Risk score, confidence level, recommended remediation steps, suggested playbook, incident severity tag.

AI continuously enriches events with context (GeoIP, user history, asset criticality) to reduce false positives.

Incident Workflow Timeline



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LOW INOWI						
BRIRD		8				
DOO7	1				98	
OOOUW	10				X	84
GUIGD						
IAW	1	1			83	
FCWW	2	8	14		33	
PAGKE						
LOW	10	3	18		63	8
ORITD=		05	06	68	06	
NISH						

Incident Management

01

Automatic Incident Creation

Detected threats create incidents with metadata and required actions.

02

Tracking & Assignment

Queue routing, role-based assignment, SLA tracking.

03

Severity Lifecycle

Open → Assigned → In Progress → Resolved → Dismissed with audit trail.

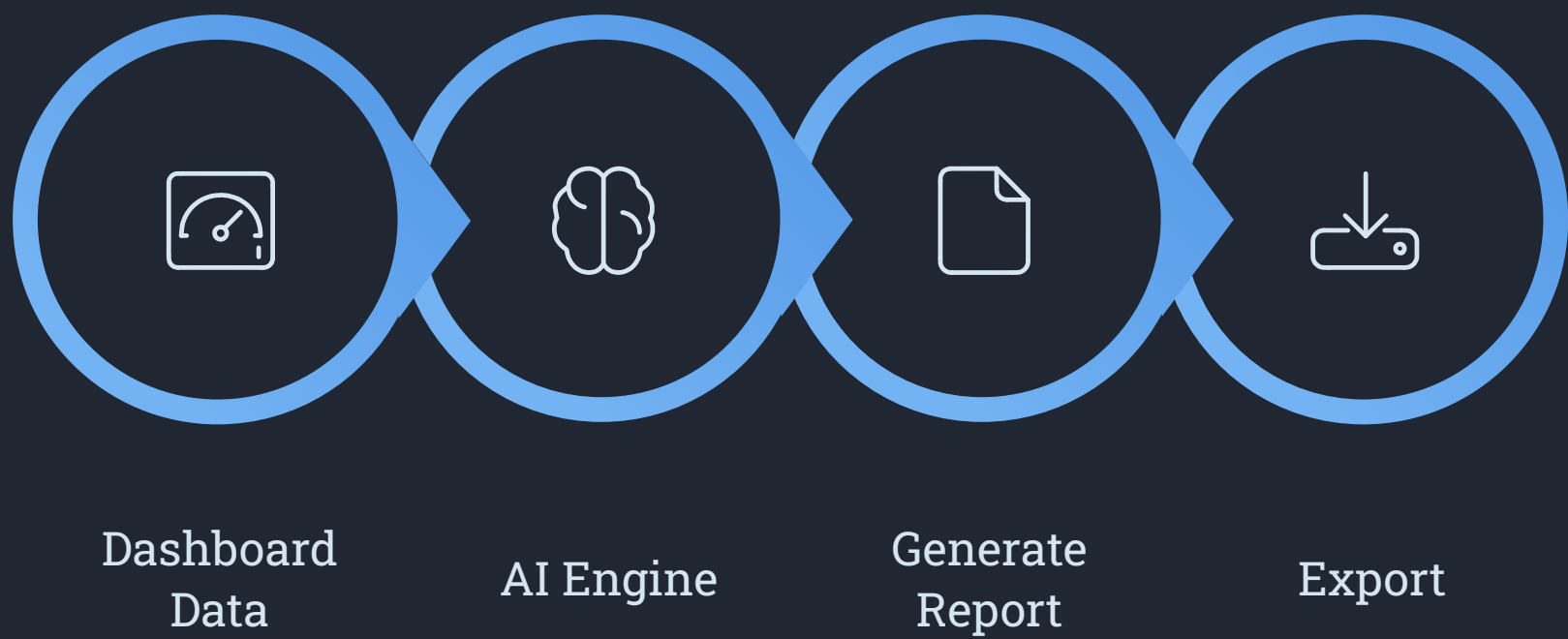
04

Priority Matrix

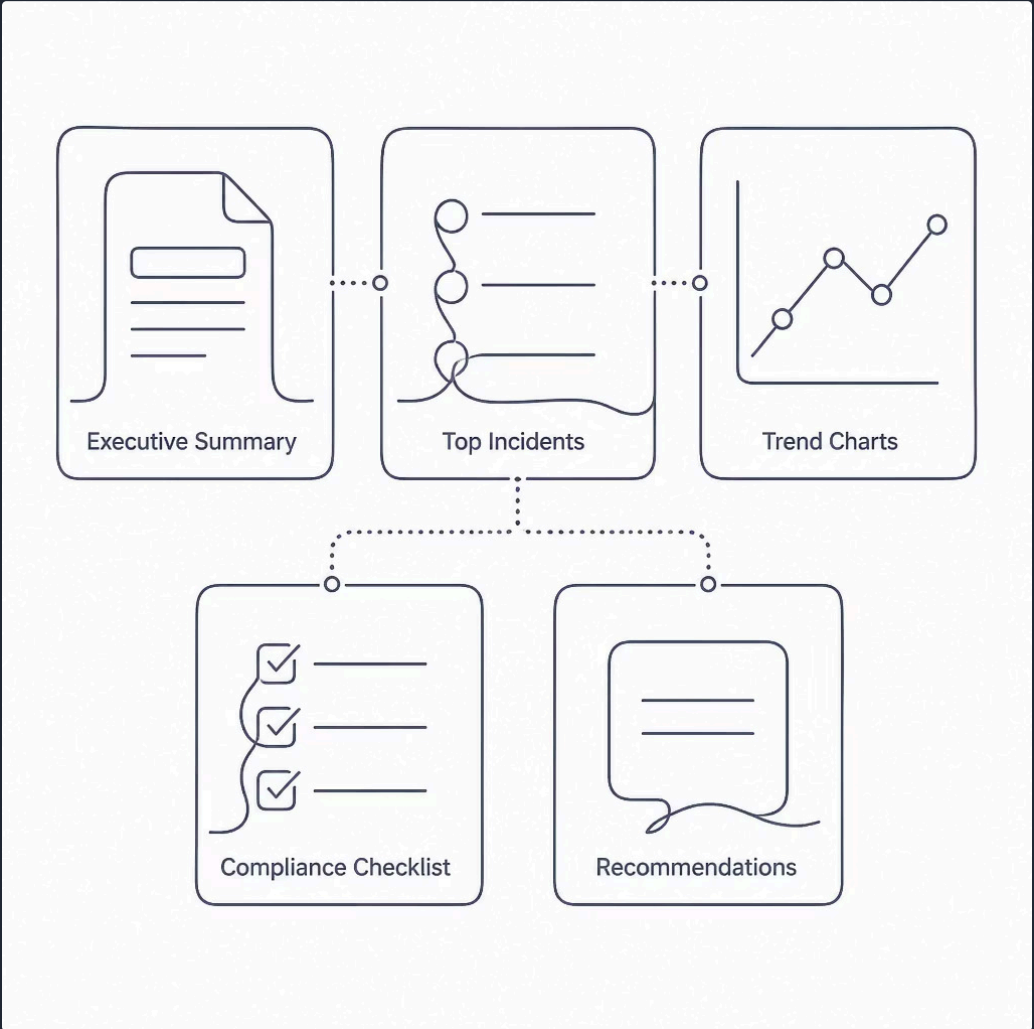
Impact vs likelihood drives response urgency and automated playbooks.

Integrated notifications (email/SNS/SMS) and evidence collection for forensic review.

AI Report Generation

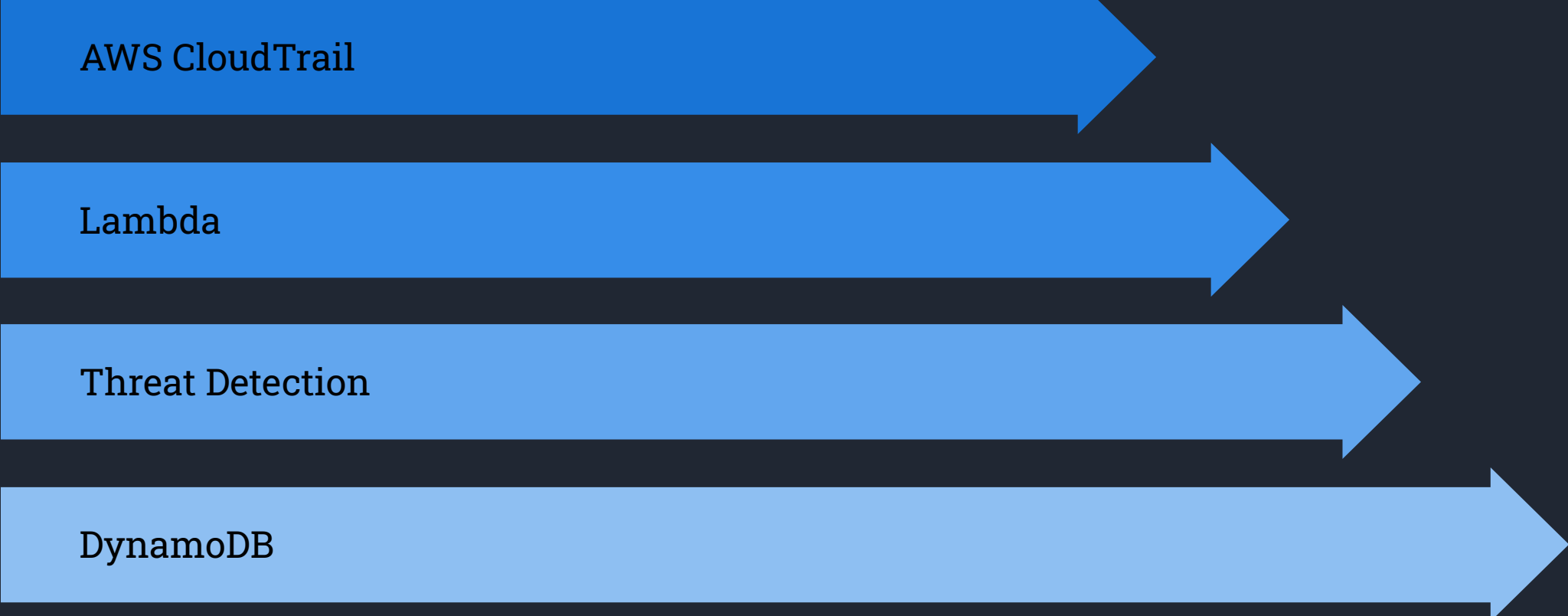


Reports include date filters, threat filters, executive summary, charts, compliance sections, and remediation recommendations. Exports: PDF, Word, TXT, Markdown.



Automated schedule options and one-click share for stakeholders and auditors.

Project Workflow (End-to-End)



Sequence emphasizes low-latency ingestion, serverless processing, persistent event storage, fast API access, and responsive dashboarding for administrators.

Conclusion & Future Scope

Conclusion

Centralized cloud SOC providing real-time monitoring, AI-assisted detection, automated incident response, and professional reporting for campus environments.

Future Scope

- ML threat prediction & anomaly forecasting
- SIEM integrations and multi-cloud coverage
- Mobile SOC app, automated remediation & playbooks
- Email & SMS alerts, compliance dashboarding

THANK YOU

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